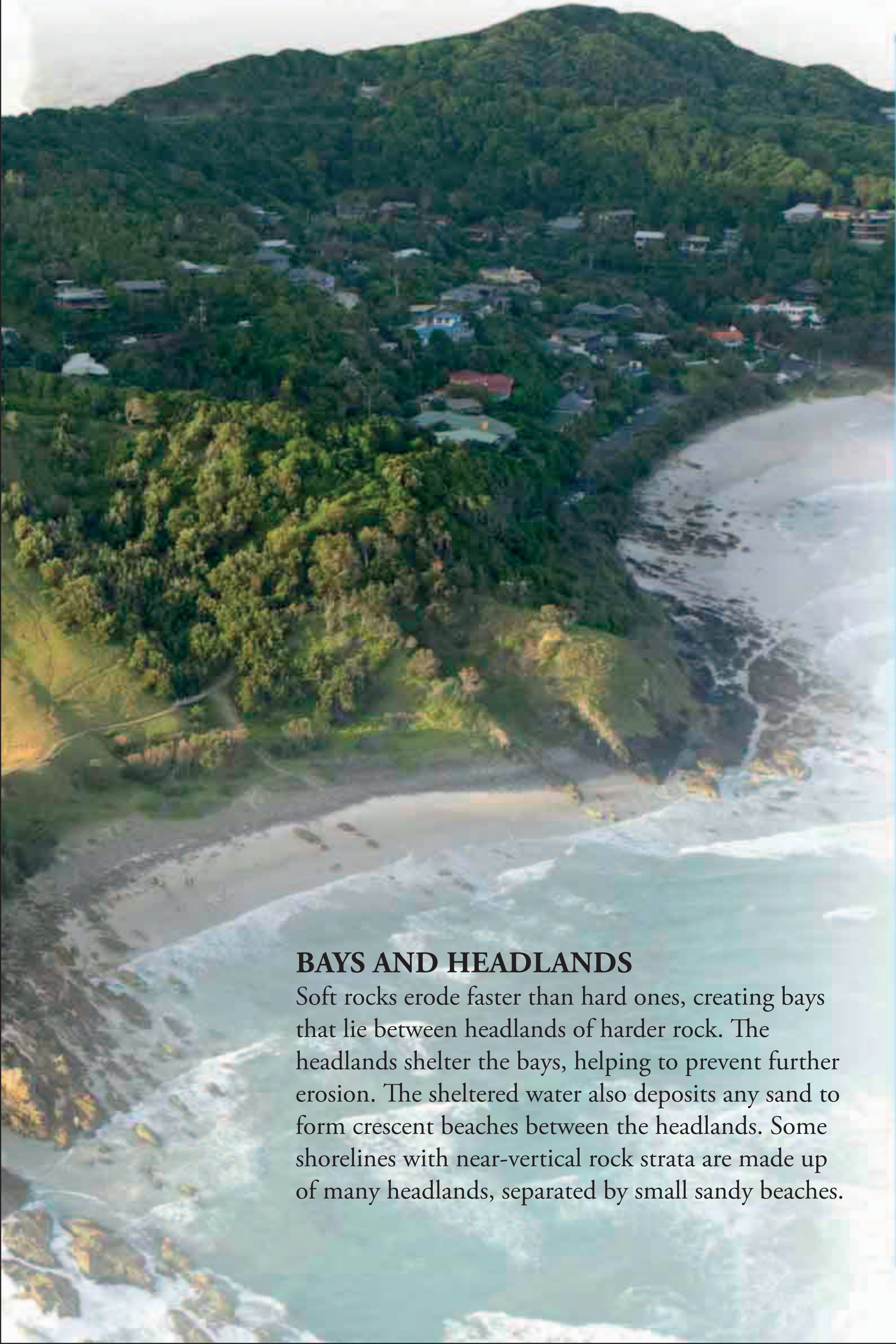
EXTENDING THE LAND

When waves break on a beach at an angle, they loosen sand and stones so that they roll into the sea. The waves pick them up and toss them back on the beach farther down the shore, so the sand and stones are steadily moved down the coast. Over time, this process can build long beaches that extend out to sea as spits.

CONSTRUCTION SITE ▶
Waves have pushed beach material from left to right along this shore in South Island, New Zealand, extending it into a long, curving sand spit.



CLIFFS, CAVES, STACKS

On some coasts, the shoreline is cut back at a steady rate to create sheer cliffs. These rise above flat platforms of rock that extend far out to sea beneath the waves. But often the coast is made of different types of rock, and the waves cut these away at different rates. The weaker rocks may then collapse while others survive, creating caves, rock arches, and isolated stacks.

▲ CARVED AWAY
Soft rock tends to crumble, creating cliffs (top). Harder rock survives undercutting to form caves and arches (middle), which often collapse to leave stacks (bottom).

BAYS AND HEADLANDS

Soft rocks erode faster than hard ones, creating bays that lie between headlands of harder rock. The headlands shelter the bays, helping to prevent further erosion. The sheltered water also deposits any sand to form crescent beaches between the headlands. Some shorelines with near-vertical rock strata are made up of many headlands, separated by small sandy beaches.